**The Hong Kong University of Science and Technology (Guangzhou)**
**Assignment 2 Continuous Random Variables**
**Please submit the soft copy of your homework solutions to Canvas**
**Due at 11:59pm on Mar 20, 2026**

1. (16 pts) The pdf (probability density function) of $X$ is given by

$$f_X(x) = \begin{cases} -\frac{1}{4}x, & -2 \le x < 0 \\ cx^2, & 0 \le x \le 3 \\ 0, & otherwise \end{cases}$$

   (a) Find the value of the constant $c$.
   (b) Plot the pdf of $X$ from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e., where they intercept the $y$-axis and $x$-axis.
   (c) Find the expression for the cdf (cumulative distribution function) of $X$.
   (d) Find the expected value of $X$.
   (e) Find the probability of the event $A = \{1 \le x \le 2\}$.
   (f) Find the conditional pdf of $X$ given $A$, $f_{X|A}(x|A)$ and plot it from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e. where they intercept the $y$-axis and $x$-axis.
   (g) Find the conditional expected value of $X$ given $A$, $E[X|A]$.

2. (16 pts) Suppose we have two machines. The lifetime of Machine 1 in years, $X_1$, follows an exponential distribution with parameter $\lambda_1 = \frac{1}{9}$. The lifetime of Machine 2 in years, $X_2$, follows an exponential distribution with parameter $\lambda_2 = \frac{1}{5}$.
   (a) Compute the expected values of $X_1$ and $X_2$. Which machine lasts longer on average?
   (b) Find the conditional pdf of $X_2$ given that the lifetime of Machine 2 is less than 4 years.
   (c) Compute the conditional expected value of $X_2$ given that the lifetime of Machine 2 is less than 4 years.
   (d) Find the probability that the lifetime of Machine 1 exceeds 9 years. Repeat for Machine 2.
   (e) Find the probability that Machine 1 will continue to work for at least another 6 years, given that the machine has been working for 9 years already.

3. (18 pts) The joint pdf of $X$ and $Y$ is given by:

$$f_{X,Y}(x,y) = \begin{cases} c(x+y)e^{-y}, & -y \le x \le y, 0 \le y \le \infty \\ 0, & otherwise \end{cases}.$$

For this question, it may be useful to draw the region in the $X$, $Y$ plane where the pdf is non-zero to help you determine the limits of the integrals.

(a) Find the value of the constant $c$.
(b) Find the marginal pdfs of $X$ and $Y$, respectively.
(c) Find the probability that $X > 2$ and $Y > 2$, $P[X > 2 \cap Y > 2]$.
(d) Find the conditional pdf of $X$ given $Y$, $f_{X|Y}(x|y)$. Plot $f_{X|Y}(x|y)$ for $y = 2$.
(e) Find the conditional expectation of $X$ given $Y = y$, $E[X|y]$. Plot $E[X|y]$ for $y$ from 0 to 6.

4. (12 pts) Let $X$ and $Y$ be jointly Gaussian random variables with means $E[X] = 1$ and $E[Y] = -4$, variances $\mathrm{Var}[X] = 4$ and $\mathrm{Var}[Y] = 9$. Define a random variable $Z = -2X + Y$.
   (a) Assume that $X$ and $Y$ are uncorrelated. Find the pdf of $Z$, $f_Z(z)$ for $-\infty < z < \infty$.
   (b) Assume that $X$ and $Y$ have a correlation coefficient $\rho = -0.3$. Find the pdf of $Z$, $f_Z(z)$ for $-\infty < z < \infty$.

5. (8 pts) Assume that the monthly salaries of Hong Kong residents are i.i.d random variables with unknown mean, but assume the standard deviation is 15,000HKD. Alice wants to use the sample mean to estimate the mean value of the monthly salaries of Hong Kong residents.
   (a) Use the Central Limit Theorem to approximate the number of Hong Kong residents that should be surveyed, $n$, such that the sample mean is within 1000HKD of the actual mean with probability greater than or equal to 96%.
   (b) Suppose that the true mean monthly salary of a Hong Kong resident is 18,000HKD. Suppose we average the monthly salary of 500 Hong Kong residents. What is the probability that the measured average is within 1000HKD of the true mean?